

# Impact Statement Card: RAG Document Retrieval For Agronomic Advice

Farmers often have questions regarding crop production, diseases, pests, soil management, and fertilizer use. Agricultural manuals, though useful, may be outdated, expensive, inaccessible, or too time-consuming to search through. This project aims to make reliable agricultural information easier, faster, and more accessible to smallholder farmers and others who need agricultural information.

To address this problem, a Retrieval-Augmented Generation (RAG) document retrieval system was developed. The system searches a collection of agricultural extension materials and identifies the most relevant documents for a farmer's question. It focuses on twelve common African crops and ranks the top five documents that are most relevant to each query. The project has the potential to reduce the time users spend searching for useful information and make existing agricultural knowledge easier to access. The system could benefit not only farmers, but also agricultural workers, students, researchers, and other agricultural practitioners.

However, the system also has some limitations and possible risks. If the knowledge base contains outdated, incomplete, or inaccurate information, the system may retrieve information that could lead to poor decisions. There is also a risk that users may trust the retrieved information without checking the original source or seeking advice from qualified agricultural professionals. In addition, the system may not perform equally well for all crops, topics, or ways of asking a question, particularly when information about certain crops is less represented in the knowledge base.

To improve the project's impact, future work should focus on expanding the knowledge base, regularly updating the document collection, improving retrieval accuracy, evaluating the system with real agricultural users, and making the sources of retrieved information clear to users. The goal is not to replace agricultural experts, but to make reliable agricultural knowledge easier to find and use, especially for people who may have limited access to agricultural extension resources.